



Royal University of Phnom Penh

Faculty of Engineering



Department: IT Engineering

Major: Data Science and Engineering

Course: Data Analytics

Lecturer: Mr. Chap Chanpiseth

Class: DSE Cohort 1

Team Members: Phork Norak, Nhor Povketya

Final Assignment Report

Continuing from our Assignment 1, our business conceptual model focuses on “HomeBuild Retail,” a family-owned small business specializing in essential construction materials and tools. This traditional retail store primarily serves walk-in customers but is exploring to add digital service to expand its reach through an e-commerce website. The store offers a comprehensive range of products, including hammers, nails, work shoes, safety equipment, lamps, showers, and various construction tools, catering to the needs of local customers. Additionally, HomeBuild Retail maintains simple customer records, collecting only phone numbers, and provides two payment options—cash and Bakong digital payments—to enhance customer convenience.

The following Figure 1 is a simple entity relationship design that represents some of the ideas from our conceptual HomeBuild Retail business model.

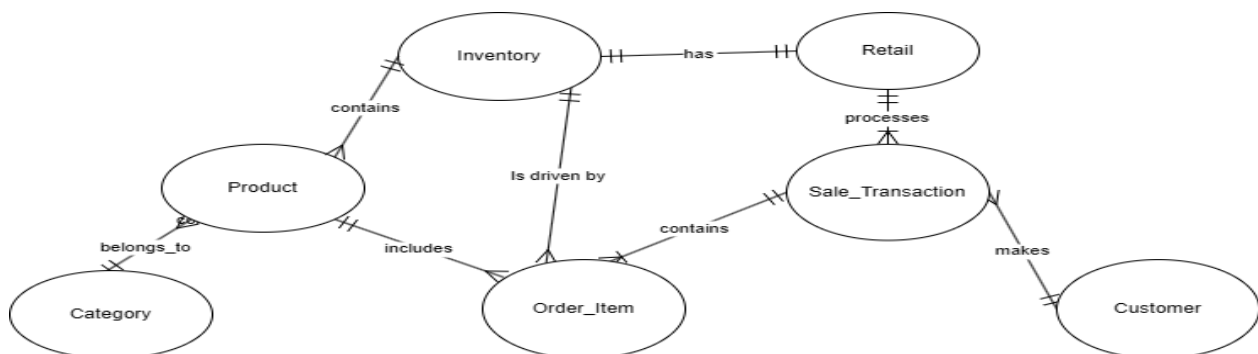


Figure 1: Entity Relationship Design

I. Introduction

The objective of this final assignment report is to show the data analytics procedure for HomeBuild Retail, a family-owned small business selling construction materials and tools, Power BI.

In this report, we cover the following key points:.

- Importing data into Power BI and establishing relationships between tables.
- Developing DAX measures to calculate key business metrics.
- Designing and implementing a comprehensive Power BI dashboard with detailed visualizations to support decision-making.

This analysis focuses on understanding customer payment behaviors, transaction patterns, and market readiness for digital transformation, addressing the critical business question: **Should HomeBuild Retail add digital services to its traditional business?**

II. Data Overview

1. Datasets

The analysis utilizes three primary datasets:

- **customers.csv**: Contains customer details, primarily phone numbers for identification.
- **products.csv**: Contains product details, including names, categories, and pricing information.
- **sale_transactions.csv**: Contains transaction details, including payment methods, quantities, and total amounts.

2. Data Descriptions

- **customers.csv**: Includes customer_id, name, contact.
- **products.csv**: Includes product_id, product_name, category_id, buy_price_per_unit, sell_price_per_unit.
- **sale_transactions.csv**: Includes transaction_id, customer_id, transaction_date, payment_method, total_amount.
- **order_item.csv**: Includes item_id, transaction_id, product_id, quantity, unit_price.
- **category.csv**: Includes category_id, category_name
- **inventory.csv**: Includes inventory_id, product_id, quantity_in_stock.

III. Data Import and Preparation

1. Import Process

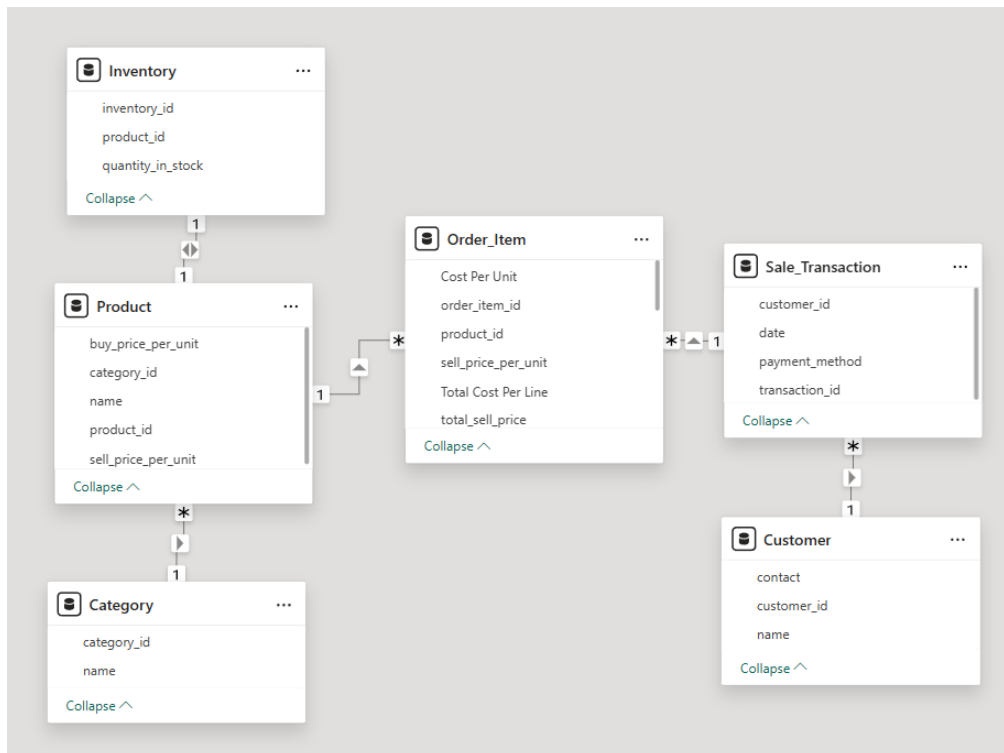
- **Import Data into Onedrive:** The CSV files (customers.csv, products.csv, sale_transactions.csv, order_item.csv, category.csv, inventory.csv) were loaded into a Onedrive folder to enable efficient querying and management.
- **Import Data into Power BI:** Using Power BI's "Get Data" function, the Onedrive file were imported to create visualizations and establish relationships for analysis.

3. Relationship Establishment

Relationships were established in Power BI to enable comprehensive analysis:

- **Sale_Transaction** linked to **Customer** via customer_id (one-to-many).
- **Order_Item** linked to **Sale_Transaction** via transaction_id (one-to-many).
- **Order_Item** linked to **Product** via product_id (many-to-one).
- **Product** linked to **Category** via category_id (many-to-one).
- **Inventory** linked to **Product** via product_id (one-to-many).

These relationships ensure that transactional data can be analyzed by customer, product, category, inventory status, and payment method dimensions.



IV. Creating Measures

2. Measures

We created a measures table in Power BI containing six DAX measures to enable dynamic calculations and aggregations for interactive visualizations:

A. Revenue

- ★ Total Revenue = SUM(Order_Item[total_sell_price])
- ★ Average Order Value = DIVIDE([Total Revenue],DISTINCTCOUNT(Order_Item[transaction_id]))

B. Profit

- ★ Total Profit = [Total Revenue] - [Total Cost]
- ★ Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)
- ★ Total Cost = SUM(Order_Item[Total Cost Per Line])
- ★ Cost Per Unit = RELATED(Product[buy_price_per_unit])
- ★ Total Cost Per Line = [unit] * [Cost Per Unit]

C. Transactions

- ★ Total Transactions = DISTINCTCOUNT(Order_Item[transaction_id])
- ★ Average Transaction Value = [Total Revenue] / [Total Transactions]

D. Units Sold

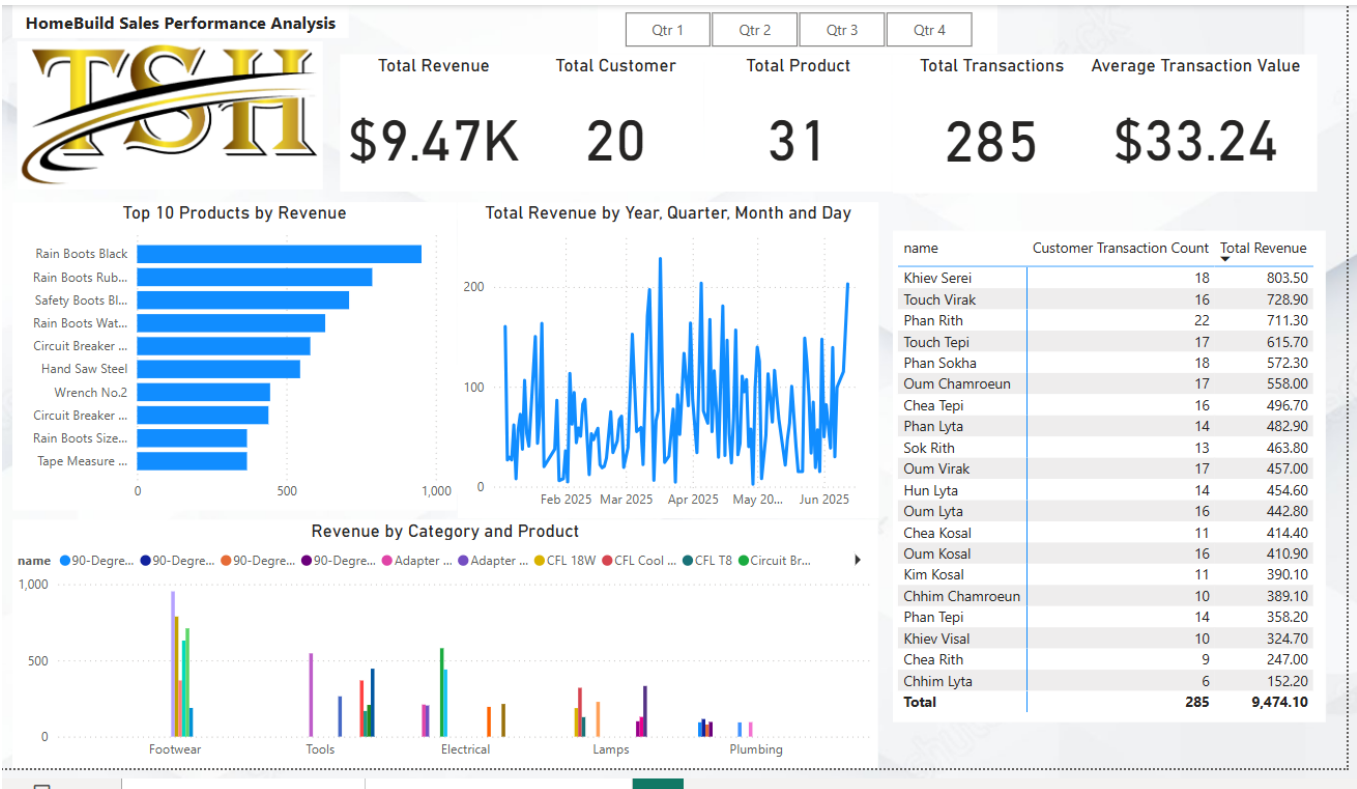
- ★ Total Units Sold = SUM(Order_Item[unit])
- ★ Total Units Sold PY =
CALCULATE([Total Units Sold],SAMEPERIODLASTYEAR('Sale_Transaction'[date]))

E. Customer Behavior

- ★ Number of Customers = DISTINCTCOUNT(Order_Item[customer_id])
- ★ Customer Transaction Count = COUNT(Sale_Transaction[transaction_id])
- ★ Repeat Customer Count =
CALCULATE(COUNTROWS(Sale_Transaction), ALLEXCEPT(Customer, Customer[customer_id]))
- ★ Customer Total Revenue =
CALCULATE(SUM(Order_Item[total_sell_price]),ALLEXCEPT(Customer, Customer[customer_id]))

V. Data Visualization in Power BI

Sales Performance Analysis Dashboard Purpose



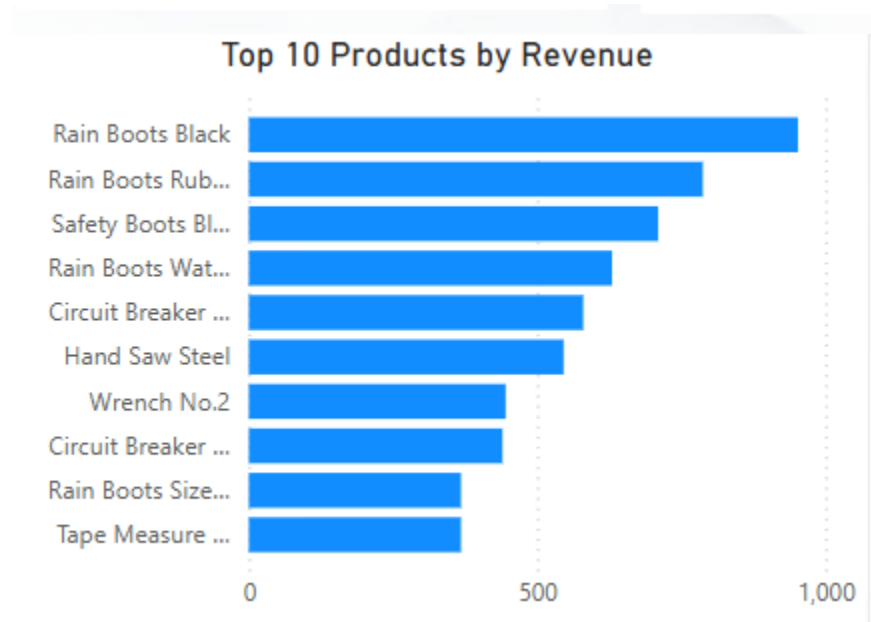
This dashboard provides a comprehensive overview of business performance from January to June 2025. The analysis highlights strong customer loyalty, clear category leaders, and consistent sales activity, with actionable insights for strategic growth in key areas such as footwear, customer retention, and product bundling

1. KPI

Total Revenue	Total Customer	Total Product	Total Transactions	Average Transaction Value
\$9.47K	20	31	285	\$33.24

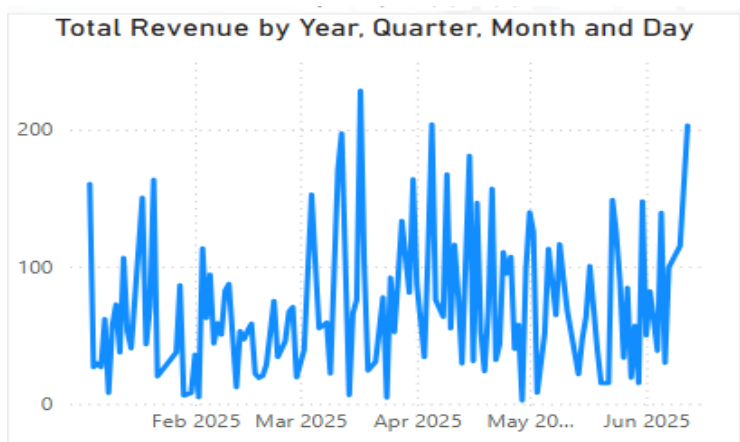
- The store made **\$9,474.10** in sales. That means it is earning money regularly.
- There were **285 sales**. This shows that people are buying often.
- On average, each sale was about **\$33.24**, which is a good amount.
- The store had **20 customers**, and many of them bought more than once.
- A total of **31 products** were sold. This means the store offers a small but useful group of items.

2. Top 10 Products by Revenue



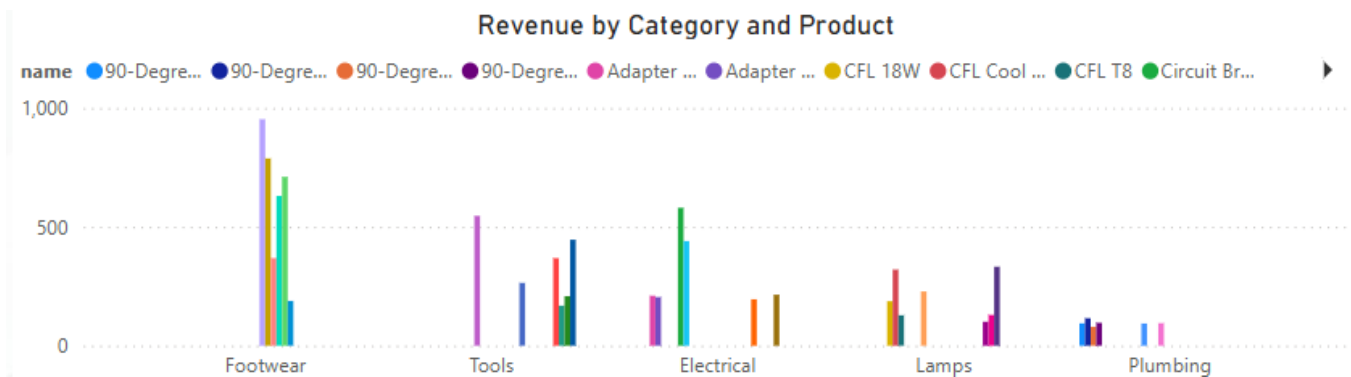
- Most of the money comes from selling **boots**, especially **Rain Boots Black**, which is the top product.
- Other best-sellers include **Rubber Boots**, **Safety Boots**, and **tools** like **Circuit Breakers** and **Hand Saws**.
- This shows that customers **mostly buy protective footwear and basic construction tools**.

3. Sales Trends Over Time



- While sales happen almost every day, the amount of money earned each day changes a lot.
- Some days the store makes less than \$50, while other days it earns more than \$200.
- From April to June (Q2), sales are higher and have more big spikes compared to January to March (Q1).
- This shows that business gets better in the second quarter of the year.

3. Category Breakdown



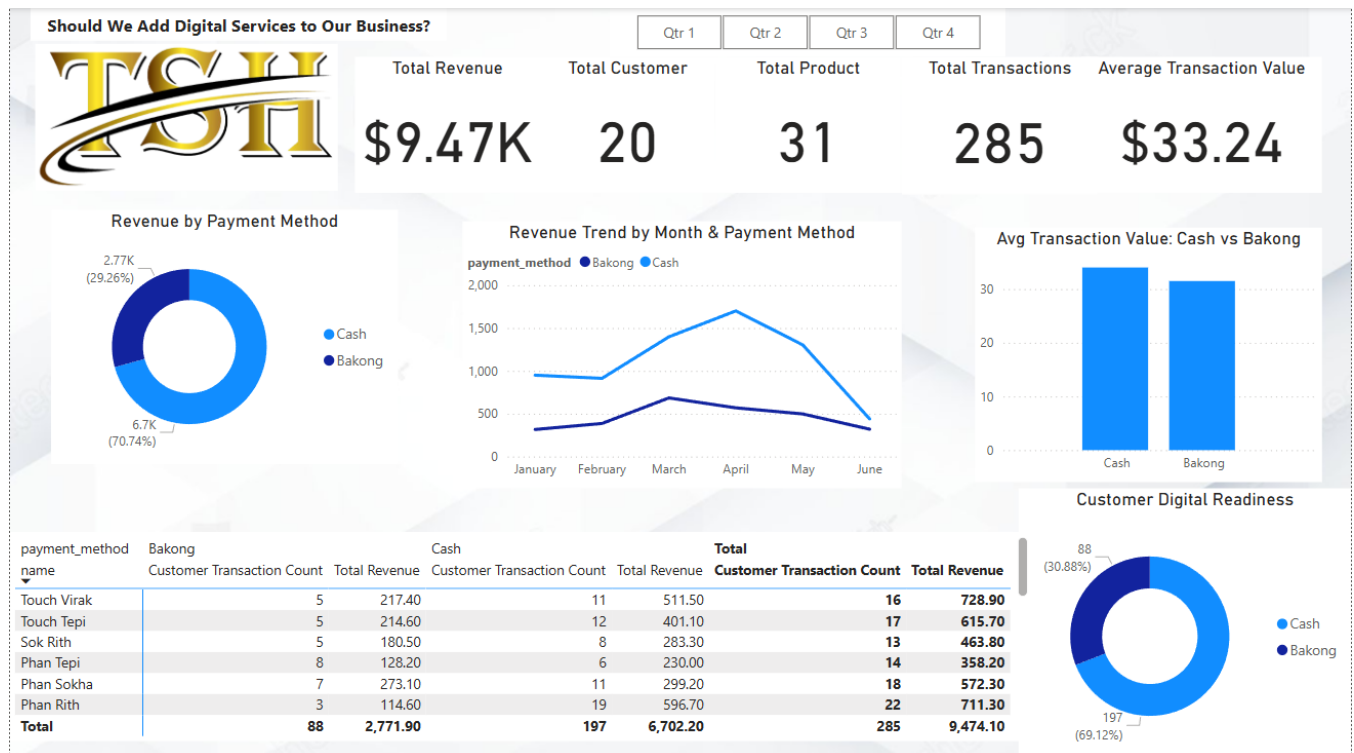
- Most of the money comes from the **Footwear** category. Products in this group, like boots, make the highest sales.
- **Tools** and **Electrical items** also earn a good amount, but not as much as Footwear.
- **Lamps** and **Plumbing** products make the least revenue.
- This shows that **protective shoes and basic tools** are the most popular items customers buy.

4. Customer Performance

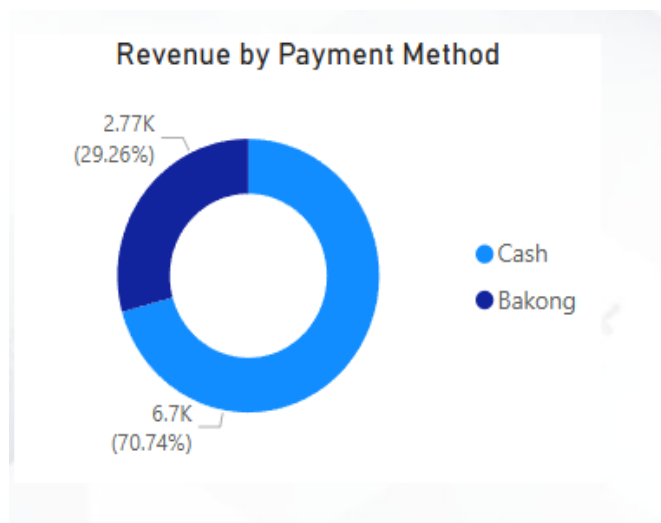
name	Customer Transaction Count	Total Revenue
Khiev Serei	18	803.50
Touch Virak	16	728.90
Phan Rith	22	711.30
Touch Tepi	17	615.70
Phan Sokha	18	572.30
Oum Chamroeun	17	558.00
Chea Tepi	16	496.70
Phan Lyta	14	482.90
Sok Rith	13	463.80
Oum Virak	17	457.00
Hun Lyta	14	454.60
Oum Lyta	16	442.80
Chea Kosal	11	414.40
Oum Kosal	16	410.90
Kim Kosal	11	390.10
Chhim Chamroeun	10	389.10
Phan Tepi	14	358.20
Khiev Visal	10	324.70
Chea Rith	9	247.00
Chhim Lyta	6	152.20
Total	285	9,474.10

- Some customers buy from the store many times.
- **Phan Rith** made the most purchases (22 times), and **Khiev Serei** spent the most money (\$803.50).
- Most top customers have **15 or more transactions**, which shows they come back often.
- This means the store has **loyal customers who shop regularly**.

Digital Transformation Decision Dashboard



1. Revenue by Payment Method

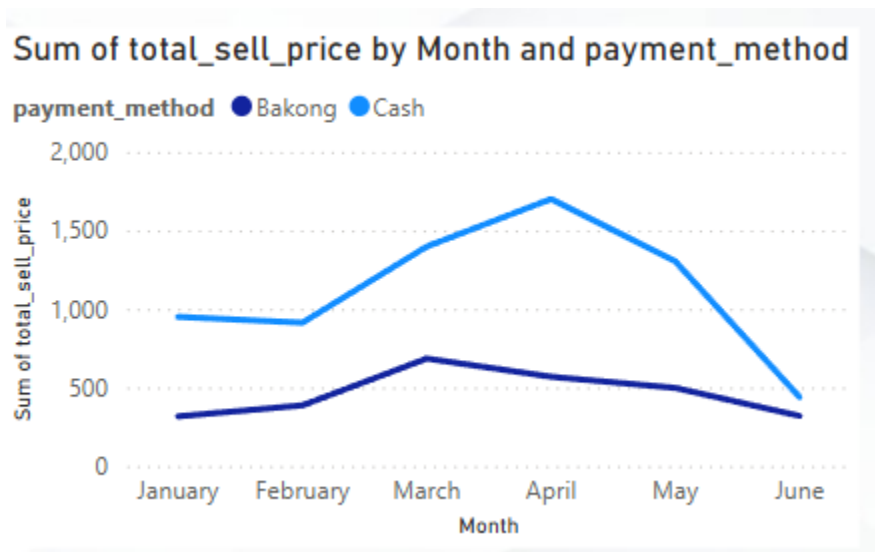


Most of the store's income comes from **cash payments**.

- **Cash** brought in about **\$6,700**, which is **70.74%** of total revenue.
- **Bakong** (digital payment) brought in **\$2,770**, or **29.26%**.

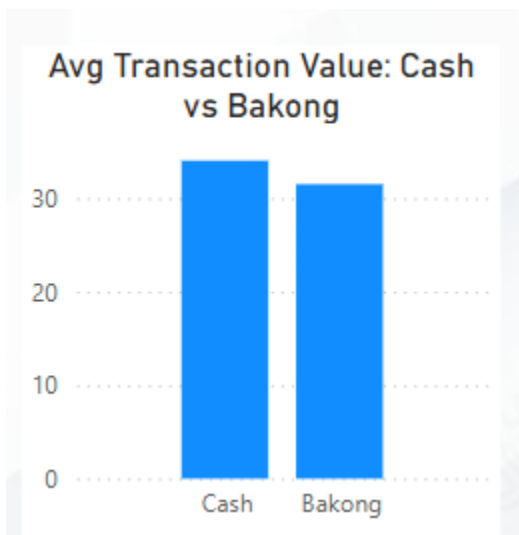
This means that **more than two-thirds** of customers still use cash, but **nearly one-third** are already using digital payment.

2. Revenue Trend by Month & Payment Method



- Cash revenue **peaks in April**, then drops
- Bakong also grows steadily until May
- Both methods follow similar seasonal patterns. Digital adoption is not limited to low-activity months

3. Avg Transaction Value: Cash vs Bakong



- **Cash spend a bit more** than those who use **Bakong**.
The **average cash transaction is about \$34**, while **Bakong is around \$31.50**.
- This means **digital payments are mostly used for smaller purchases**, but they are still growing.

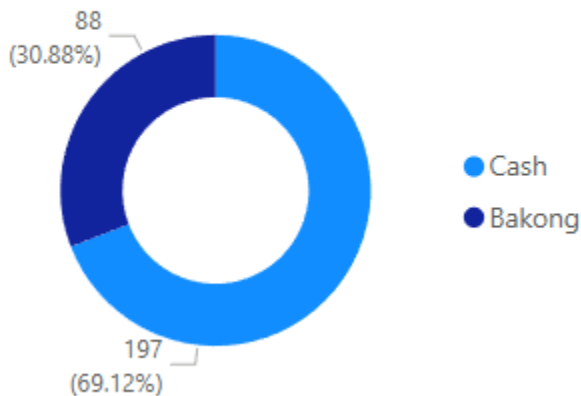
4. Comparison of top customers using Cash vs Bakong

payment_method name	Bakong		Cash		Total			
	Customer	Transaction Count	Total Revenue	Customer	Transaction Count	Total Revenue	Customer Transaction Count	Total Revenue
Touch Virak		5	217.40		11	511.50	16	728.90
Touch Tepi		5	214.60		12	401.10	17	615.70
Sok Rith		5	180.50		8	283.30	13	463.80
Phan Tepi		8	128.20		6	230.00	14	358.20
Phan Sokha		7	273.10		11	299.20	18	572.30
Phan Rith		3	114.60		19	596.70	22	711.30
Total		88	2,771.90		197	6,702.20	285	9,474.10

- Top customers use **both Cash and Bakong**.
- **Phan Rith** made 22 purchases, using both methods, and spent over **\$700 total**.
- Most revenue still comes from **cash payments**, but **Bakong is used regularly**, especially by loyal customers.
- This means customers are **comfortable switching between payment types**, and digital use can grow with more support.

5. Customer Digital Readiness

Customer Digital Readiness



- **30.88% of transactions were made using Bakong**, while **69.12% used cash**.
- This means that **about 1 in 3 customers are already using digital payments**, showing a good level of digital readiness.

VI. Result

Our Power BI dashboards provide clear and evidence-based results:

- **Revenue** totaled **\$9,474.10** over 6 months.
- **Protective footwear** accounts for the top-selling products, especially rain boots.
- **30.88% of customers** already use **Bakong** digital payments.
- **Cash remains dominant** but both payment methods follow similar seasonal patterns.
- **Digital users spend 7.4% less per transaction** compared to cash users.
- **Most top customers use both payment methods**, showing digital convenience without abandoning traditional behavior.

These findings support a **hybrid business model**. We recommend that HomeBuild Retail begin integrating **digital services**, to **enhancing customer convenience**. Initial steps may include a **simple e-commerce site, loyalty program**, or **online pre-order system**, especially for frequently purchased products like footwear and tools.

VII. Conclusion

This report demonstrates how data analytics using Power BI can help HomeBuild Retail evaluate its performance and make strategic decisions. Through a series of connected datasets and visual dashboards, we explored customer behavior, product performance, payment patterns, and digital adoption trends.

The **Sales Performance Analysis** dashboard reveals that HomeBuild generated approximately **\$9,474** in revenue from **285 transactions** between January and June 2025. The top-selling products are mostly **footwear**, especially rain and safety boots, indicating high demand for protective gear in the construction sector. Sales activity was consistent, with **peak performance during March to May**, and customer loyalty was evident 20 customers generated 285 transactions, with several making over 15 purchases.

The **Digital Service Investment** dashboard explores whether HomeBuild should expand into digital channels. While **cash remains dominant** (70.7% of revenue), **Bakong digital payments account for 29.26% of revenue and 30.88% of transactions**, indicating moderate digital readiness. However, **cash transactions average slightly higher** per order than Bakong (\$34.02 vs. \$31.50), suggesting that digital users are more value-conscious rather than high spenders.

The insights suggest that HomeBuild has a **stable core business** with loyal, repeat customers and a clear product focus. There is also a strong base of digital adopters. Therefore, a gradual move toward **e-commerce or digital service features** such as online ordering, product reservations, or digital invoices is supported by the data.